

CHAPTER 6

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- 6.1. (a) Let M be a Noetherian A -module and $u : M \rightarrow M$ a module homomorphism. If u is surjective, then it is an isomorphism.
(b) If M is Artinian and u is injective, then again u is an isomorphism.

Proof. (a) Consider the filtration $\ker u \subset \ker u^2 \subset \cdots$. M Noetherian gives us $n \geq 1$ such that $\ker u^n = \ker u^{n+r}$ for all $r \geq 0$. Then $\square$

- 6.4. Let M be a Noetherian A -module, and let $\mathfrak{a}$ be the annihilator of M in A . Prove that $A/\mathfrak{a}$ is a Noetherian ring.

If we replace *Noetherian* by *Artinian*, is this result still true?

- 6.11. Let $f : A \rightarrow B$ be a ring homomorphism and suppose that $f^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ is a closed mapping if and only if f has the going up property (5, Ex. 10).